

TenderHub AI Agent Subsystem: Specifications & Continuation Plan

Technical Roadmap for Contractor Intelligence and Tender Dossier Engine

Author: Lead Systems Architect Date: October 2026 Classification: Internal Technical Manual Version: 2.4.0

ABSTRACT

Detailed implementation runbook for the incoming engineer continuing development of the Contractor AI Subsystem. Covers multi-tier fallback mechanics, Cloudflare R2 PDF text extraction, prompt structures, and token cost economics.

1. AI Subsystem Scope & Objectives

The Contractor AI Agent subsystem transforms unstructured government tender specifications (NIT PDFs) into structured engineering intelligence answering qualification thresholds, BOQ material breakdowns, and post-award execution milestones.

2. Multi-Tier Architecture & Fallback Mechanics

Tier Level	Engine / Provider		Activation Condition	Output Fidelity
Tier 1	Google Gemini 1.5 Flash	<code>GEMINI_API_KEY</code>	valid and configured	Full contextual extraction with large context
Tier 2	OpenAI <code>gpt-4o-mini</code>		Gemini fails or key missing; OpenAI key present	JSON extraction using chat completion format
Tier 3	Deterministic Engine	Rule	Zero external API keys or rate-limit exceptions	Calculated civil engineering quantities & milestones

3. Step-by-Step Technical Instructions for Continuation

Step 1: Configure Live Gemini Credentials

Generate a key in Google AI Studio and configure in `backend/.env`:

```
GEMINI_API_KEY=AIzaSy... your_key_here ...
GEMINI_MODEL=gemini-1.5-flash
```

Step 2: Stream PDF Document Text from Cloudflare R2

In `backend/src/services/contractorAgent.service.js`, implement document retrieval:

```
import { GetObjectCommand } from '@aws-sdk/client-s3';
import { s3Client } from '../config/r2.js';
import pdfParse from 'pdf-parse';

async function extractTextFromTenderDocument(tender) {
  const docKey = tender.nitDocumentKey || tender.documents?.[0]?.s3Key;
  if (!docKey) return '';
  const command = new GetObjectCommand({
    Bucket: process.env.R2_BUCKET_NAME || 'tenderhub',
    Key: docKey,
  });
  const response = await s3Client.send(command);
  const chunks = [];
  for await (const chunk of response.Body) chunks.push(chunk);
  const buffer = Buffer.concat(chunks);
  const pdfData = await pdfParse(buffer);
  return pdfData.text.slice(0, 150000);
}
```

Step 3: Enable Context Caching for Cost Optimization

Use the Google AI Context Caching API to cache ingested tender documents for 24 hours. Repeat requests from different contractors will cost \$0.01875 per 1M cached tokens (75% discount) and respond in under 2 seconds.

4. Economic Analysis

- Cost per 30-page tender dossier analysis: ~\$0.003 (~₹0.25 INR)
- Pro AI Tier customer price: ₹999.00 / month (10 Dossiers)
- Compute expense per subscriber: ~₹15.00 INR
- **Gross Margin: > 98%**